Decision-Density Evaluation Allocation Strategy for Decomposition-Based MOEA:Supplementary Material*

Han Zhang(✉), Cheng Wang, Cheng Liu, and Huangxing Lin

College of Computer Science and Technology, Huaqiao University, Xiamen, Fujian 361021, China

24014083065@stu.hqu.edu.cn,{wangcheng,liucheng,linhuangxing}@hqu.edu.cn

1 Description of the test question

Multiobjective Order Fulfillment Problem(OF): In the second set of experiments, a mathematical model of the actual order fulfillment problem is established for testing. Given n independent orders, each order contains demand for up to m types of products. There is a distributed inventory system composed of k different warehouses with no direct connections among them. Each warehouse has different initial inventory levels and processing capacities for various products. The core decision problem is to allocate one or more supply sources from k candidate warehouses for up to m product demand items of n orders, while optimizing three objectives: order fulfillment rate, profit margin, and inventory utilization. This is a type of multi-objective combinatorial optimization problem.

1.1 Decision Variable

$$\mathbf{y_0^w(g)} \in \{\mathbf{0}, \mathbf{1}\} \tag{1}$$

Where $y_o^w(g)$ is defined as a three-dimensional binary decision variable. If the value of $y_o^w(g)$ is 1, it means that product g in order o is supplied by warehouse w; if the value of $y_o^w(g)$ is 0, it means that product g in order o is not supplied by warehouse w.

1.2 Objective Function

$$f = \begin{cases} max\left(\frac{\Sigma_{o=1}^{|O|}\sum_{g=1}^{|G|} s_o(g)p_o(g)}{\Sigma_{o}^{|O|}\sum_{g=1}^{|G|} p_o(g)}\right) \\ max\left(\frac{\sum_{o=1}^{|O|}\sum_{g=1}^{|G|} s_o(g)}{\Sigma_{o=1}^{|O|}\sum_{g=1}^{|G|} 1}\right) \\ max\left(\frac{\sum_{o=1}^{|O|}\sum_{g=1}^{|G|} s_o(g)d_o(g)}{\Sigma_{g=1}^{|G|}\sum_{w=1}^{|W|} s^w(g)}\right) \\ max\left(-\frac{TVC_{max}-\sum_{G(g)=1}^{|G|} VC_{G(g)}}{TVC_{max}-TVC_{min}}\right) \end{cases} \tag{2}$$

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The specific meaning of objective functions is explained below.

1.3 Constraint Conditions

$$\sum_{o=1}^{|O|}\sum_{w=1}^{|W|} aw_o^w(g) \leq \sum_{o=1}^{|O|} \Sigma_{g=1}^{|G|} d_o(g) \tag{3}$$

$$aw_o^w(g) \leq \min(d_o(g), s^w(g)) \tag{4}$$

$$0 \leq \Sigma_{o=1}^{|O|} \Sigma_{g=1}^{|G|} \Sigma_{w=1}^{|W|} y_o^w(g) \leq c^w(g) \tag{5}$$

Equation (3) represents the inventory allocation constraint, which means the total quantity allocated for a certain product in an order cannot exceed its demand. Here, $aw_0^w(g)$ denotes the total quantity allocated for a certain product in an order. Equation (4) represents the warehouse inventory capacity constraint, which means the total quantity of products allocated from a warehouse cannot exceed its inventory, and the inventory level must be greater than 0. Equation (5) represents the warehouse processing capacity constraint, which means the number of orders for each product allocated from a warehouse cannot exceed its processing limit $c^w(g)$.

2 Evaluation metrics of Experiment 2

Given a set of obtained non-dominated solutions NDS and a reference point set P^*, the IGD is defined as:

$$IGD(\mathbf{NDS}, \mathbf{P}^*) = \frac{1}{|\mathbf{P}^*|} \sum_{\mathbf{x}^* \in \mathbf{P}^*} d\left(\mathbf{x}^*, \mathbf{NDS}\right) \tag{6}$$

Where d denotes the Euclidean distance from each reference point $x^* \in P^*$ to its nearest solution x in the NDS. In the experiments, the reference point set P^* is constructed by uniformly sampling 10,000 points from the true PF of each benchmark problem. A smaller IGD value indicates better performance.

HV requires a reference point r, and computes the sum of the hypervolumes formed by the solution set NDS and r. The HV value is defined as:

$$HV(\boldsymbol{NDS}, r) = \mathcal{L}\left(\bigcup_{x_a \in \boldsymbol{NDS}} \{\boldsymbol{x}_b | \boldsymbol{x}_a < \boldsymbol{x}_b < r\}\right) \tag{7}$$

Where $\mathcal{L}$ denotes the Lebesgue measure, representing the volume of the union of all points x_b that are dominated by the reference point r and dominate any solution $x_a \in P$. The solutions are normalized in the objective space, and the reference point r is set to 1 in all dimensions (e.g., for a 3-objective problem, $r = (1.0, 1.0, 1.0)$). A larger HV value indicates better performance.

In the second set of experiments, beyond IGD and HV, the actual problem evaluation metrics include profit margin C, order fulfillment rate N, inventory utilization U, and order priority violation degree TVC, defined as follows:

$$C = \frac{\Sigma_{o=1}^{|O|}\Sigma_{g=1}^{|G|}s_o(g)p_o(g)}{\Sigma_{o}^{|O|}\Sigma_{g=1}^{|G|}p_o(g)} \tag{8}$$

$$N = \frac{\Sigma_{o=1}^{|O|}\sum_{g=1}^{|G|}s_o(g)}{\Sigma_{o=1}^{|O|}\sum_{g=1}^{|G|}1+\varepsilon} \tag{9}$$

$$U = \frac{\Sigma_{o=1}^{|O|}\Sigma_{g=1}^{|G|}s_o(g)d_o(g)}{\Sigma_{g=1}^{|G|}\Sigma_{w=1}^{|W|}s^w(g)} \tag{10}$$

Where o and g denote the indices of orders and goods, respectively; $|O|$ and $|G|$ represent the total number of orders and types of goods; $s_o(g)$ is an indicator function that equals 1 if the demand for product g in order o is fulfilled, and 0 otherwise; $p(g)$ denotes the profit associated with product g in order o. This formulation underpins the calculation of the profit margin C, reflecting the economic return from successfully fulfilled order items. $\varepsilon = 1e-6$. $d_o(g)$ represents the demand quantity of product g in order o, and $s^w(g)$ represents the inventory level of product g in warehouse w.

$$TVC = \frac{TVC_{\max} - \sum_{G(g)=1}^{|G|}VC_{G(g)}}{TVC_{\max} - TVC_{\min}} \tag{11}$$

$$VC_{G(g)} = G_i(g)max(G_{NO,o}(g) - GF_{NO,o}(g), 0) \tag{12}$$

Where $TVC_{\max}$ and $TVC_{\min}$ denote the theoretical maximum and minimum penalty values, respectively; $VC_{G(g)}$ is the penalty value for the product group $G(g)$; i represents the priority level of $G(g)$; $G_i(g)$ indicates whether product g in group $G(g)$ satisfies priority i; $G_{N0,o}(g)$ denotes the arrival sequence number of product g in order o within group $G(g)$; and $GF_{N0,o}(g)$ represents the sequence number of the first unfilled product in group $G(g)$.

3 Experimental Results And Analysis

Due to the large scale and practical nature of the problem, the true PF is intractable. Thus, the true PF for this problem is set as the PS obtained by each algorithm with a computational budget of 80,000/800,000.The computational resources for the 2-objective and 3-objective experiments are 40000, and those for the 5-objective experiment are 400000.

Table S1. presents the mean and standard deviation of HV values obtained by 6 MOEAs after 30 independent runs on different benchmark problems with varying numbers of objectives. The results show that MOEA/D-DDEA outperforms the other comparison algorithms. Among the 21 test groups, MOEA/D-DDEA achieves the best HV value in 47.61% of the cases, MOEA/D-BOS in 28.57%, MOEA/D-ENS in 14.28%, HK-MOEAD in 4.76%, and NSGA- in 4.76%. In terms of statistical tests, MOEA/D-DDEA performs better in 80.76% of the 104

comparisons. Notably, the proposed algorithm outperforms the other comparison algorithms in the 2-objective and 3-objective scenarios of DTLZ4 and DTLZ7, indicating that the algorithm maintains good performance on problems with nonlinear and irregular PFs. Although the algorithm does not show a significant advantage in the 2-objective ZDT problems, it still achieves promising results.

The results in Fig. S1 indicate that the solution set obtained by MOEA has the best diversity and uniformity.In Fig. S2, MOEA/D-DDEA evidently achieves a high approximation to the discontinuous and irregular true PF of DTLZ7, with good uniformity as well.

Fig. S3 shows the final solution distribution of MOEA/D-DDEA on various benchmark problems. MOEA/D-DDEA demonstrates excellent high-dimensional multi-objective optimization capability on the DTLZ series of test problems. The solution set generated by the algorithm is uniformly and densely distributed within the corresponding cubic objective space, and can well fit the Pareto fronts with different topological structures such as convex and spherical shapes for different test problems, fully reflecting the algorithm's adaptability and coverage ability to complex high-dimensional objective spaces.

The algorithm also performs impressively on the ZDT series of test problems. For classic convex-front problems like ZDT1 and ZDT2, the solution set generated by the algorithm presents a continuous curve distribution from the upper left corner to the lower right corner, which highly matches the true front, and the solutions are uniformly distributed on the curve without discontinuity or local aggregation. For the multimodal test problem ZDT3, the algorithm successfully captures the multimodal characteristics of its Pareto front, with uniformly distributed solutions on each segment of the discontinuous curve, proving the algorithm's effectiveness in handling multimodal multi-objective optimization problems. Moreover, for high-dimensional complex problems like ZDT4 and ZDT6, the algorithm can also cover the entire Pareto front, further verifying its optimization capability in high-dimensional complex scenarios.

From Fig. S4 , it can be observed that as K increases, the HV value increases, but when K is 50% of the population size, the HV value drops sharply. The optimal K value is 25% of the population size, which performs better than 10% in both overall performance and the maximum HV value. Therefore, this experiment demonstrates that setting the K value to 25% of the population size yields better results.

Table S1. HV values of six algorithms in DTLZ and ZDT benchmark problems. The best result for each test case is shown in bold.

MOPS	M	NSGA-III	MOEA/D	HK-MOEA/D	MOEA/D-ENS	MOEA/D-BOS	MOEA/D-DDEA
DTLZ1	2	8.6340e-1	8.1535e-1	8.2361e-1	5.8260e-1	8.5304e-1	**8.7319e-1**
		(4.09e-3)-	(9.07e-2)-	(1.30e-3)-	(3.55e-8)-	(1.19e-3)-	**(1.59e-3)**
	3	8.1147e-1	2.3736e-1	8.4163e-1	**8.4440e-1**	7.3017e-1	6.5738e-1
		(1.85e-1)+	(2.82e-1)-	(7.71e-5)+	**(2.72e-6)+**	(2.02e-1)+	(3.04e-1)
	5	9.9455e-1	9.3443e-1	9.7042e-1	9.7493e-1	9.5007e-1	**9.9517e-1**
		(1.15e-3)≈	(2.77e-2)-	(2.90e-4)-	(1.35e-4)-	(1.82e-2)-	**(1.15e-3)**
DTLZ2	2	6.2597e-1	4.6211e-1	5.0743e-1	3.4721e-1	4.9239e-1	**6.3268e-1**
		(4.79e-3)-	(4.78e-2)-	(3.75e-2)-	(6.29e-8)-	(3.81e-2)-	**(1.17e-3)**
	3	5.9325e-1	5.0745e-1	5.5955e-1	5.6302e-1	**7.4918e-1**	6.6231e-1
		(5.32e-2)-	(2.46e-2)-	(2.94e-5)-	(1.76e-5)-	**(1.70e-2)+**	(1.35e-2)
	5	9.0729e-1	8.2135e-1	7.7371e-1	7.9380e-1	**9.0991e-1**	9.0594e-1
		(5.23e-3)≈	(7.53e-2)-	(5.10e-4)-	(4.25e-4)-	**(2.06e-2)+**	(2.86e-3)
DTLZ3	2	9.0697e-1	8.8973e-1	9.2505e-1	9.4672e-1	9.9405e-1	**9.9936e-1**
		(5.39e-2)-	(1.96e-2)-	(1.04e-1)-	(1.04e-2)-	(6.78e-3)-	**(3.39e-4)**
	3	7.5953e-1	7.5915e-1	5.4914e-1	5.6272e-1	**9.4289e-1**	8.5500e-1
		(7.35e-2)-	(8.97e-2)-	(7.15e-3)-	(2.68e-4)-	**(2.77e-2)+**	(3.20e-2)
	5	7.0100e-1	7.5201e-1	7.3700e-1	7.9031e-1	**9.8183e-1**	9.6871e-1
		(2.24e-2)-	(1.61e-2)-	(5.64e-2)-	(1.34e-3)-	**(8.64e-3)+**	(3.87e-3)
DTLZ4	2	4.5081e-1	3.7499e-1	4.5083e-1	3.4721e-1	5.2869e-1	**6.0347e-1**
		(1.51e-1)-	(3.22e-6)-	(1.51e-1)-	(9.84e-8)-	(1.77e-1)-	**(1.52e-1)**
	3	6.8057e-1	4.7868e-1	5.4460e-1	5.6302e-1	5.4646e-1	**8.4260e-1**
		(1.04e-1)-	(6.97e-3)-	(5.65e-2)-	(3.02e-5)-	(1.28e-1)-	**(5.05e-2)**
	5	6.4726e-1	7.7554e-1	7.4551e-1	**7.9384e-1**	6.7449e-1	6.7266e-1
		(3.55e-2)-	(7.86e-2)+	(6.49e+2)-	**(5.72e-4)+**	(1.32e-1)≈	(3.78e-2)
DTLZ7	2	3.0329e-1	2.6261e-1	2.0579e-1	2.4253e-1	2.9327e-1	**3.0606e-1**
		(1.77e-5)-	(5.51e-2)-	(1.02e-1)-	(2.04e-7)-	(6.29e-3)-	**(2.76e-3)**
	3	2.9967e-1	2.3578e-2	2.7142e-1	2.6583e-1	2.0917e-1	**3.1351e-1**
		(1.65e-1)-	(1.42e-2)-	(7.02e-4)-	(1.17e-4)-	(1.84e-1)-	**(1.01e-1)**
	5	-	4.3276e-3	6.9013e-3	1.4734e-1	**2.7657e-1**	7.0591e-2
			(4.81e-3)-	(7.17e-3)-	(2.31e-4)+	**(1.74e-2)+**	(1.97eS3)
ZDT1	2	5.1957e-1	4.7499e-1	4.8716e-1	**7.2031e-1**	7.1960e-1	6.1993e-1
		(2.93e-3)-	(2.78e-2)-	(8.55e-2)-	**(2.59e-6)+**	(2.45e-4)+	(1.08e-5)
ZDT2	2	4.9895e-1	1.6486e-1	1.6580e-1	4.4503e-1	4.4453e-1	**5.1013e-1**
		(5.47e-3)-	(9.84e-4)-	(3.75e-4)-	(3.13e-7)-	(2.57e-4)-	**(9.93e-6)**
ZDT3	2	**7.7405e-1**	7.1627e-1	7.4476e-1	5.9998e-1	6.1748e-1	7.6890e-1
		(9.47e-3)+	(4.95e-2)-	(2.84e-2)-	(3.12e-6)-	(4.03e-2)-	(1.94e-3)
ZDT4	2	6.3445e-1	7.5958e-1	**8.4757e-1**	7.2030e-1	7.1895e-1	7.1216e-1
		(1.04e-1)-	(7.60e-3)+	**(6.68e-2)+**	(3.04e-6)+	(8.65e-2)+	(1.88e-1)
ZDT5	2	4.7034e-1	4.3403e-1	4.2174e-1	4.5182e-1	4.3890e-1	**4.8647e-1**
		(1.46e-2)-	(6.81e-2)-	(1.23e-2)-	(1.55e-2)-	(1.36e-2)-	**(1.87e-2)**
ZDT6	2	6.4419e-1	3.2395e-1	3.4030e-1	2.7698e-1	**7.1852e-1**	6.7136e-1
		(1.07e-1)-	(2.30e-2)-	(2.34e-2)-	(4.66e-2)-	**(5.92e-5)+**	(7.20e-4)

Table S2. Results of 6 algorithms on the order fulfillment problem.Where OSI denotes the number of orders in which all item demands are fulfilled, GSI denotes the number of fulfilled item entries in each order, and $\lambda = 0.1$.

Algorithm	M	OSI	GSI	U	N	C	$\lambda * TVC$	IGD	HV
NSGA-III	2	77	1053	3.1734e-1	4.5512e-1	-	-	8.1764e-1	4.8161e-1
	3	87	981	2.8969e-1	4.2423e-1	2.3921e-1	-	5.6053e-1	3.0644e-2
	4	-	-	-	-	-	-	-	-
MOEAD	2	87	1010	3.1954e-1	4.3732e-1	-	-	8.5451e-1	4.7392e-1
	3	85	938	2.8284e-1	4.0602e-1	2.1084e-1	-	5.5321e-1	2.2357e-1
	4	76	978	3.0881e-1	4.2300e-1	2.3445e-1	1.7514e-2	8.6431e-1	**7.1827e-1**
HK-MOEAD	2	84	1067	3.1799e-1	4.6131e-1	-	-	9.3800e-1	4.8416e-1
	3	85	1034	3.1445e-1	4.4780e-1	2.3739e-1	-	8.1840e-1	3.3861e-2
	4	91	1058	3.0720e-1	4.5871e-1	2.3162e-1	1.7854e-2	8.3604e-1	5.3731e-1
MOEA/D-ENS	2	92	1085	3.1629e-1	4.6973e-1	-	-	5.4062e-1	4.8482e-1
	3	83	1093	3.1773e-1	4.7319e-1	2.4918e-1	-	6.5582e-1	3.0377e-1
	4	81	1033	**3.2293e-1**	4.4767e-1	2.4334e-1	**1.6692e-2**	**3.3337e-1**	3.9110e-1
MOEAD-BOS	2	83	1055	3.2152e-1	4.5609e-1	-	-	**4.2307e-1**	4.8115e-1
	3	**93**	1030	3.1393e-1	4.4571e-1	2.2990e-1	-	5.3278e-1	2.9275e-1
	4	**93**	**1099**	3.1338e-1	**4.7559e-1**	**2.4891e-1**	1.7719e-2	3.7873e-1	1.9289e-1
MOEA/D-DDEA	2	**102**	**1147**	**3.3930e-1**	**4.9640e-1**	-	-	7.1470e-1	**5.0172e-1**
	3	**95**	**1145**	**3.2321e-1**	**4.9539e-1**	**2.5148e-1**	-	**4.6343e-1**	**4.5324e-1**
	4	83	999	3.0553e-1	4.3244e-1	2.2029e-1	1.7223e-2	4.0181e-1	3.3291e-1

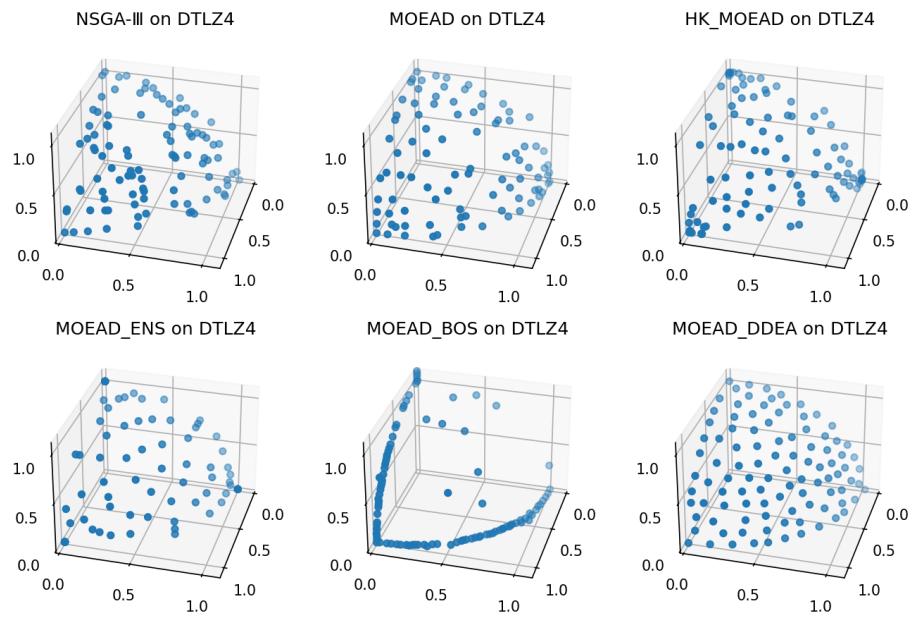

Fig. S1. Solution set distribution of 6 algorithms on the DTLZ4 benchmark problem

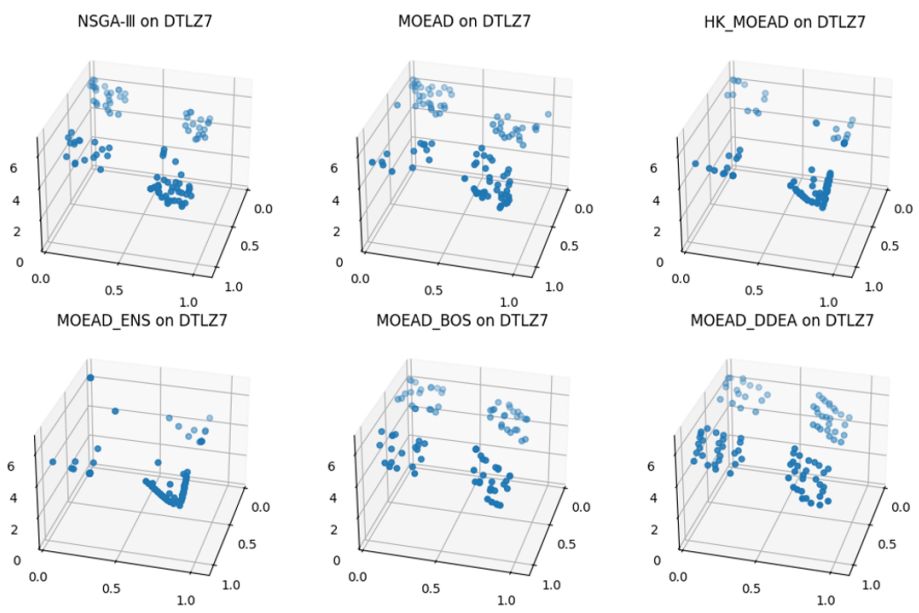

Fig. S2. Solution set distribution of 6 algorithms on the DTLZ7 benchmark problem

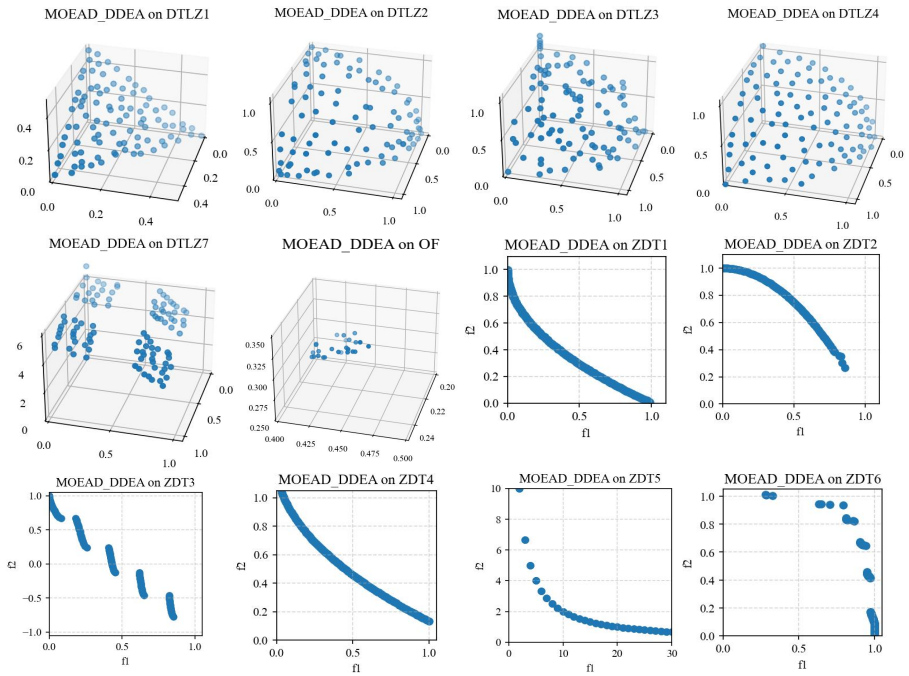

Fig. S3. Solution set distribution of MOEA/D-DDEA on various benchmark problems

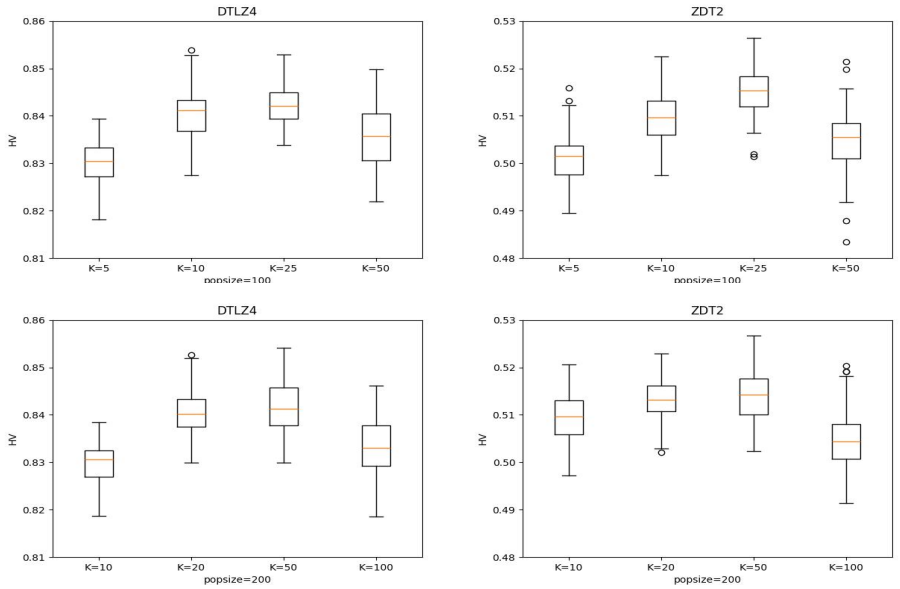

Fig. S4. The effect of K on HV values